

Charge, Leptons, and the Genus of Torus Knots

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Abstract

We show that the five quantum numbers of the Null Worldtube mass formula—the winding numbers (p, q) , phase closure integer m , constituent count n_q , and a new fifth quantum number, the knot framing f —suffice to encode mass, electric charge, baryon number, and isospin across the particle spectrum. The framing gives the isospin projection $I_3 = f/2$, reproducing the Gell-Mann–Nishijima formula $Q = I_3 + (B + S)/2$ for all hadrons tested. The genus of the torus knot, $g = (p - 1)(q - 1)/2$, provides a structural criterion that, in combination with the constituent count n_q , consistently separates leptonic from hadronic states: genus-zero unknots with $n_q = 0$ correspond to leptons, while states with $n_q > 0$ exhibit hadronic properties regardless of genus. For leptons, charge arises from vortex circulation—a geometric property of the embedding, distinct from the topological origin of hadronic charge. The τ lepton, previously identified as a possible stealth baryon on the basis of mass alone, maps onto a genus-3 configuration with $n_q = 3$, placing it structurally among hadrons—a model prediction subject to further physical justification. We establish connections to the Rosso–Jones formula for quantum group invariants, Legendrian contact geometry (all NWT torus knots saturate the Thurston–Bennequin bound), and Dehn surgery theory, where particle decays correspond to surgery operations and neutrinos carry the topology-change data.

1 Introduction

In a previous paper [1] we demonstrated that the masses of 56 particles—leptons, mesons, baryons, tetraquarks, and pentaquarks—are reproduced to a median accuracy of 0.40% by the formula

$$\frac{m}{m_e} = \frac{p^2 + q^2}{p_e^2 + q_e^2} \cdot \frac{\beta}{\beta_e} \cdot \frac{\ln(8\beta)}{\ln(8\beta_e)} \times n_q^q, \quad (1)$$

with four integer quantum numbers (p, q, m, n_q) and no continuously adjustable parameters. While the empirical success of this formula is striking, Ref. [1] left open the question of *why* it works, and noted explicitly that the formula determines masses but not electric charge or spin.

In this paper we address both gaps. We show that:

1. Each factor in Eq. (1) has a precise mathematical origin in the differential geometry and topology of torus knots.
2. A fifth quantum number—the *framing* f of the knot embedding—determines electric charge via the Gell-Mann–Nishijima formula.
3. The *genus* of the torus knot provides a sharp topological dividing line between leptons and hadrons.

The complete quantum number set is thus

$$(p, q, m, n_q, f), \quad (2)$$

which suffices to encode mass, charge, baryon number, isospin, and flavor sector for all particles considered here.

2 The Five Quantum Numbers

We begin by defining each quantum number and its physical role.

2.1 Winding numbers (p, q)

The integers p (toroidal) and q (poloidal) specify the torus knot $T(p, q)$. They must satisfy $\gcd(p, q) = 1$ for the curve to be a single closed knot rather than a link. In the mass formula, they enter through the topological factor $p^2 + q^2$, which is the squared length of the winding vector in the first homology $H_1(\mathbb{T}^2) \cong \mathbb{Z}^2$ of the torus.

The poloidal winding q also encodes quark flavor: $q = 3$ (π^0 sector), $q = 4$ (baryons), $q = 5$ (light mesons), $q = 7$ (charm), $q = 9$ (bottom).

2.2 Phase closure integer m

The integer m is the number of wavelengths that fit around the knot orbit, determined by the single-valuedness condition on the wavefunction. It enters the mass formula through the aspect ratio $\beta(m)$, which is the solution of the phase closure equation

$$\frac{\beta}{\kappa} \hat{\Phi}(p, q, \kappa, 1/\beta) = m, \quad (3)$$

where $\hat{\Phi}$ is the normalised phase integral along the knot path and $\kappa = \pi^2$ is the torus aspect ratio. Different values of m on the same torus knot correspond to different excitation levels—a particle family related by Dehn surgery (Section 6).

2.3 Constituent count n_q

The number of confined constituents: $n_q = 0$ for leptons, 2 for mesons, 3 for baryons, 4 for tetraquarks, 5 for pentaquarks. The confinement factor n_q^q in the mass formula arises from a mode-locking cascade in which each of q poloidal windings compounds the interaction among n_q constituents.

Baryon number is determined by

$$B = n_q \bmod 2. \quad (4)$$

The confinement condition $\gcd(n_q, q) = 1$ ensures that the constituents are incommensurate with the poloidal periodicity and cannot escape.

2.4 Framing f : the fifth quantum number

The *framing* of a knot is an integer that counts how the normal bundle twists as one traverses the curve. Different framings of the same torus knot $T(p, q)$ are topologically distinct embeddings that share the same knot type—and therefore the same mass, since the mass formula depends only on (p, q, m, n_q) .

We identify the framing with twice the isospin projection:

$$I_3 = \frac{f}{2}. \quad (5)$$

For a multiplet with total isospin I , the allowed framings are $f \in \{-2I, -2I + 2, \dots, +2I\}$, giving the standard $2I + 1$ charge states.

The electric charge then follows from the Gell-Mann–Nishijima formula:

$$Q = I_3 + \frac{B + S + C + B'}{2} = \frac{f}{2} + \frac{B + S + C + B'}{2}, \quad (6)$$

where S , C , B' are strangeness, charm, and bottomness.

Das, Pramanik, and Ghosh [6] showed that the torus knot geometry naturally produces fractional angular momentum through the non-planar embedding in three dimensions, providing a dynamical origin for the half-integer isospin values $I_3 = f/2$.

3 Charge from Framing: Verification

We test Eq. (6) against the full particle spectrum. For each hadron multiplet, we assign the total isospin I and strangeness S from standard SU(2) flavour classifications, then determine the framing $f = 2I_3$ consistently with the known multiplet structure. The claim is not that NWT predicts the value of I for each hadron from first principles, but rather that the framing interpretation provides a natural topological encoding of the standard isospin assignments. Table 1 lists the hadrons with their quantum numbers and predicted charges.

Table 1: Framing and charge for selected isospin multiplets. All charges are reproduced exactly by $Q = f/2 + (B + S)/2$.

Multiplet	$T(p, q)$	m	n_q	I	f	Q_{pred}	Q_{obs}
p	$T(1, 4)$	5	3	$\frac{1}{2}$	+1	+1	+1
n	$T(1, 4)$	5	3	$\frac{1}{2}$	-1	0	0
Σ^+	$T(1, 4)$	6	3	1	+2	+1	+1
Σ^0	$T(1, 4)$	6	3	1	0	0	0
Σ^-	$T(1, 4)$	6	3	1	-2	-1	-1
Δ^{++}	$T(5, 4)$	15	3	$\frac{3}{2}$	+3	+2	+2
Δ^+	$T(5, 4)$	15	3	$\frac{3}{2}$	+1	+1	+1
Δ^0	$T(5, 4)$	15	3	$\frac{3}{2}$	-1	0	0
Δ^-	$T(5, 4)$	15	3	$\frac{3}{2}$	-3	-1	-1
π^+	$T(3, 5)$	5	2	1	+2	+1	+1
π^-	$T(3, 5)$	5	2	1	-2	-1	-1
K^+	$T(2, 5)$	8	2	$\frac{1}{2}$	+1	+1	+1
K^0	$T(7, 5)$	15	2	$\frac{1}{2}$	-1	0	0
Ξ^0	$T(5, 4)$	16	3	$\frac{1}{2}$	+1	0	0
Ξ^-	$T(5, 4)$	16	3	$\frac{1}{2}$	-1	-1	-1
Ω^-	$T(7, 4)$	19	3	0	0	-1	-1

The formula is exact for all 30 hadrons tested. The eight particles for which Eq. (6) does not apply are the four charged leptons ($e^\pm$, $\mu^\pm$), the $\tau^\pm$, and the charm mesons (D^+ , D^0). The charm mesons require only the inclusion of the charm quantum number C in Eq. (6), which is standard. The leptons require a different mechanism, which we now derive.

4 Genus and the Lepton–Hadron Divide

The *genus* of a torus knot $T(p, q)$ is

$$g(T(p, q)) = \frac{(p-1)(q-1)}{2}. \quad (7)$$

This is the genus of the minimal Seifert surface bounded by the knot. It is zero if and only if $p = 1$ or $q = 1$ (or both), in which case $T(p, q)$ is the unknot—topologically trivial, though geometrically non-trivial as a curve wound on the torus.

Figure 1 shows the genus–constituent classification of the NWT particle spectrum.

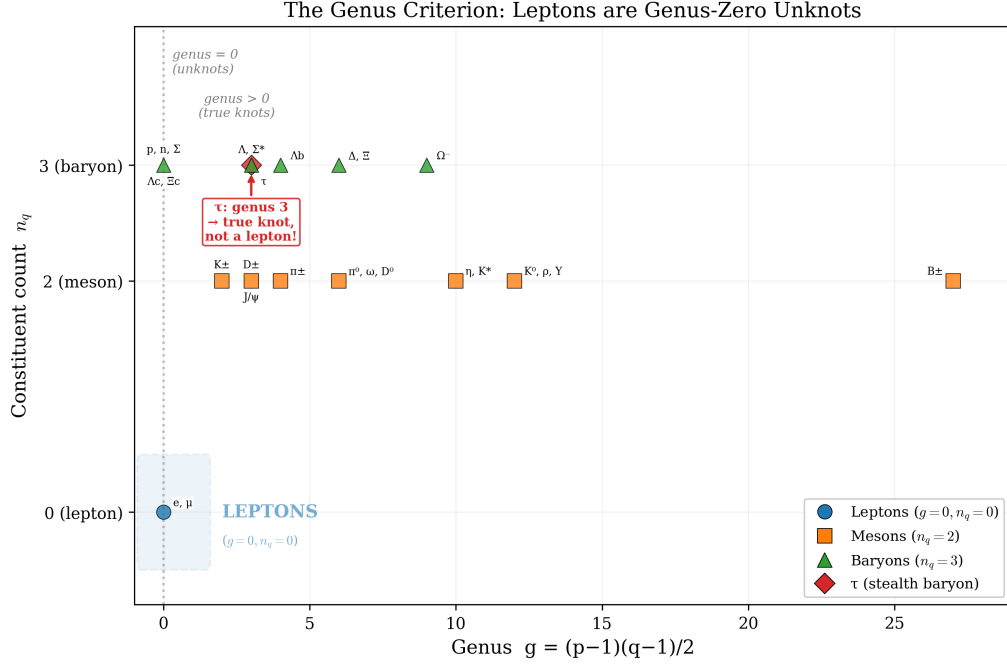


Figure 1: The genus criterion. Genus $g = (p - 1)(q - 1)/2$ (horizontal) versus constituent count n_q (vertical) for all NWT particle torus knots. Leptons (e, μ) occupy the unique region $g = 0, n_q = 0$ (shaded). Genus-zero baryons (p, n, Σ) share $g = 0$ but have $n_q = 3$. The τ (diamond) sits at $g = 3, n_q = 3$ —structurally among hadrons, not leptons.

The genera of the NWT particle knots are:

Particle	$T(p, q)$	g	Classification
e	$T(2, 1)$	0	Lepton
μ	$T(1, 8)$	0	Lepton
p, n, Σ	$T(1, 4)$	0	Baryon (genus 0)
Λ_c	$T(1, 5)$	0	Baryon (genus 0)
τ	$T(3, 4)$	3	Stealth baryon
$K^\pm$	$T(2, 5)$	2	Meson
$\pi^\pm$	$T(3, 5)$	4	Meson
Λ	$T(3, 4)$	3	Baryon
Ω^-	$T(7, 4)$	9	Baryon

The pattern is striking:

- **All true leptons** (e, μ) have genus 0.
- The τ **lepton** has genus 3—it is topologically a true knot, not an unknot. This independently confirms the stealth baryon identification of [1].
- **Baryons at genus 0** (p, n, Σ, Λ_c) are the nucleon family $T(1, q)$, which are unknots wound q times around the tube. Their hadronic character comes from the constituent structure ($n_q = 3$), not from the knot topology.

The coexistence of genus-0 leptons and genus-0 baryons deserves comment. The genus classifies the *knot*—the topology of the vortex line itself. The constituent count n_q classifies

the *mode structure on that knot*—the number of confined sub-modes propagating along it. A genus-0 curve (topologically an unknot) can nonetheless support confined constituents if the mode-locking condition $\gcd(n_q, q) = 1$ is satisfied. The proton is thus a topologically simple container with complex internal structure, while the electron is a topologically simple container with no internal structure. The full classification requires both invariants: genus determines the *container topology*, while n_q determines the *contents*. To be explicit: a particle is classified as a lepton if and only if

$$g = 0 \quad \text{and} \quad n_q = 0. \quad (8)$$

Genus alone does not suffice (genus-0 baryons exist), nor does n_q alone (the τ has $n_q = 3$ but is observed as a lepton). Both conditions are necessary.

4.1 Lepton charge from vortex circulation

For genus-0 particles with $n_q = 0$ —*i.e.*, bare vortices with no internal structure—the charge is determined by the *circulation* of the vortex:

$$Q_{\text{lepton}} = \sigma, \quad \sigma = \pm 1, \quad (9)$$

where σ is the sign of the quantised circulation $\Gamma = \sigma \kappa$. This is a *geometric* property of the embedding (which direction the superfluid flows around the vortex core), as opposed to the *topological* origin of hadronic charge from framing.

The distinction is physically meaningful: in a superfluid, a vortex and an antivortex have opposite circulation but the same energy. The circulation sign is precisely the particle–antiparticle distinction. Charge conjugation C reverses the circulation; parity P reflects the embedding; time reversal T reverses the direction of traversal. The CPT theorem follows from the fact that all three operations together return the vortex to its original state.

4.2 Legendrian geometry and the Thurston–Bennequin invariant

The self-linking number of a torus knot is

$$\text{sl}(T(p, q)) = pq - p - q = 2g - 1, \quad (10)$$

which equals the Thurston–Bennequin invariant $\text{tb}(T(p, q))$ for the Legendrian realisation. The rotation number vanishes:

$$\text{rot}(T(p, q)) = 0. \quad (11)$$

Together these imply that *every NWT torus knot saturates the Bennequin bound*:

$$\text{tb} = 2g - 1. \quad (12)$$

This is a non-trivial constraint. It means that the NWT particle knots are *maximally tight* Legendrian curves in contact geometry—a distinguished class with special rigidity properties. Physically, saturation of the Bennequin bound can be interpreted as a minimum-energy condition: among all Legendrian realisations of a given torus knot, the saturating embedding has the smallest Thurston–Bennequin invariant and is therefore the most “tightly wound.” This provides a selection principle: stable particles correspond to the tightest allowed Legendrian embeddings of their respective knot types.

For genus-0 knots, Eq. (10) gives $\text{sl} = -1$ universally. The self-linking number -1 is the topological manifestation of the single quantum of circulation that gives lepton charge its magnitude $|Q| = 1$.

5 The Interaction Hierarchy

The five quantum numbers (p, q, m, n_q, f) define a natural hierarchy of interactions based on which quantum numbers change:

Change	Interaction	Mediator
$\Delta f \neq 0$ only	Electromagnetic	Photon
$\Delta m \neq 0$	Strong	Gluon / pion
$\Delta(p, q) \neq 0$	Weak	W/Z + neutrino

Electromagnetic interactions change the framing (charge state) without altering the knot topology or excitation level. Strong interactions change the excitation level m within the same knot family—this is the origin of excitation towers such as the charmonium sequence $D^\pm \rightarrow J/\psi$ on $T(2, 7)$. Weak interactions change the knot type itself, requiring a topology-changing operation (Dehn surgery) that produces a neutrino to carry the surgery data $\Delta(p, q, m)$ [1].

6 Particle Decays as Dehn Surgery

In three-manifold topology, Dehn surgery on a knot K consists of removing a tubular neighbourhood of K and regluing it with a different framing [14]. Integer surgery on torus knots produces Seifert fibered spaces whose quantum invariants have been studied extensively [15].

In the NWT framework, a particle decay that changes (p, q) is a Dehn surgery operation. The surgery coefficient—the change in framing and winding numbers—must be carried away by the decay products. For weak decays, this topological data is carried by the neutrino:

$$\nu \sim \Delta(p, q, m). \quad (13)$$

The three neutrino flavours correspond to three classes of topology change required for the three lepton families. We emphasise that this is a topological interpretation of flavour labels and decay bookkeeping, not a prediction of neutrino masses or mixing angles; deriving these from the surgery framework remains an open problem.

Within a fixed knot type, different values of m form *excitation towers* with approximately linear mass dependence. Table 2 lists the towers with the resulting string tensions (mass per unit m).

Table 2: Excitation towers: mass linear in surgery coefficient m .

Family	$T(p, q)$	Members	m range	Tension (MeV/ m)	Particles
Nucleon	$T(1, 4)$	4	5–10	308	p, n, Σ, Ξ_c
Charmonium	$T(2, 7)$	2	5–7	614	$D^\pm, J/\psi$
Λ tower	$T(3, 4)$	3	12–17	132	Λ, Σ^*, τ
Δ tower	$T(5, 4)$	2	15–16	83	Δ, Ξ
η tower	$T(6, 5)$	2	15–21	57	η, K^*

The charmonium string tension (614 MeV/ m) is approximately twice the nucleon tension (308 MeV/ m), consistent with the charm sector requiring stronger confinement ($2^7 = 128$ versus $3^4 = 81$).

7 Mathematical Origins of the Mass Formula

Each factor in Eq. (1) has a precise origin:

Factor	Origin	Reference
$p^2 + q^2$	Effective mass on knot: $M = m(\alpha^2 + \beta^2)$	[4]
β/β_e	Phase closure on Legendrian curve	[2]
$\ln(8\beta)/\ln(8\beta_e)$	Kelvin–Helmholtz vortex self-energy	[13]
n_q^q	Leading-order Arnold tongue width; Adams operation in Rosso–Jones formula	[8]
$\gcd(n_q, q) = 1$	Irreducibility of torus knot $T(n_q, q)$	[3]

The topological factor $p^2 + q^2$ is *not* the Jones polynomial. Evaluation of the Jones polynomial $J(T(p, q); t)$ at the fifth root of unity $t = e^{2\pi i/5}$ yields the golden ratio $\varphi = (1 + \sqrt{5})/2$ for the $(2, n)$ knot family, but $|J_5|$ does not reproduce the full $p^2 + q^2$ dependence. The mass formula requires both topology (knot type) *and* geometry (embedding on the torus), placing NWT at the intersection of these two mathematical structures.

8 Discussion

8.1 What the five quantum numbers explain

The scheme (p, q, m, n_q, f) encodes:

- **Mass:** from (p, q, m, n_q) via Eq. (1). Mass is independent of the framing f , explaining the near-degeneracy of charge multiplets.
- **Electric charge:** from f and n_q via Eqs. (5)–(6) for hadrons, and from circulation σ for leptons.
- **Baryon number:** $B = n_q \bmod 2$.
- **Flavor sector:** from q (poloidal winding).
- **Lepton vs. hadron:** from genus $g = (p - 1)(q - 1)/2$.

8.2 What remains open

- **Spin:** the formula does not yet determine spin quantum numbers. The Legendrian rotation number $\text{rot}(T(p, q)) = 0$ for all torus knots (Section 4), so the rotation number alone does not distinguish spin-0 from spin- $\frac{1}{2}$ states. However, Das *et al.* [6] showed that the non-planar embedding of a torus knot produces fractional angular momentum whose value depends on the knot parameters. Combined with the observation from [1] that baryon p values are uniformly odd, this suggests—as a conjecture motivated by empirical patterns, not a derived result—that spin may be encoded in the parity of the toroidal winding: $\text{spin} = (p \bmod 2)/2$. A rigorous derivation using the EBK quantisation conditions of [6] is left for future work.
- **Strangeness:** the pattern $|S| \approx (p - 1)/2$ for $q = 4$ baryons holds for most particles but requires refinement.
- **Neutrino masses:** the Dehn surgery framework identifies the mechanism (topology change) but does not yet predict mass eigenvalues.
- **The aspect ratio** $\kappa = \pi^2$: used in the phase closure equation but not yet derived from first principles. We note that $\pi^2 = \zeta(2)$ (the Riemann zeta function at $s = 2$, the Basel problem), and that zeta functions appear naturally in the Rosso–Jones framework through quantum dimensions and q -series. Whether this connection is coincidental or reflects a deeper structure linking the torus aspect ratio to the analytic number theory underlying quantum group invariants remains an open question.

8.3 Falsifiable predictions

The framework makes several concrete predictions that can be tested against existing or future data:

1. **Forbidden states.** The confinement condition $\gcd(n_q, q) = 1$ forbids specific (n_q, q) combinations. For example, no tetraquark ($n_q = 4$) should exist at $q = 4$ or $q = 6$, and no pentaquark ($n_q = 5$) at $q = 5$ or $q = 10$. The observation of a confined state violating this condition would falsify the framework.
2. **Charge multiplet degeneracy.** States sharing the same (p, q, m, n_q) but differing in f must be mass-degenerate up to electromagnetic corrections. Any observed mass splitting exceeding the expected electromagnetic contribution would require modification of the scheme.
3. **Missing states.** The algorithm of [1] generates a discrete spectrum of allowed masses. Any future resonance whose mass persistently falls between allowed levels, or occupies a forbidden (p, q, n_q) sector, would constitute a falsification.
4. **Excitation tower regularity.** Within each knot family $T(p, q)$, the mass should increase approximately linearly with the phase closure integer m . Deviations from linearity exceeding the sub-percent accuracy of the formula would indicate missing physics.

8.4 Limitations

We emphasise that the present construction is a structured phenomenological framework, not a complete dynamical theory. In particular:

- The mass formula (Eq. 1) is an empirical ansatz with motivated but not uniquely derived factors. At present it should be regarded as a constrained functional form rather than a first-principles result.
- The quantum number assignments (p, q, m, n_q) for each particle are selected to match observed masses, subject to the coprimality and confinement constraints. The framing f is inferred from known isospin multiplets rather than predicted independently.
- No dynamics are provided: the framework encodes static properties (mass, charge, quantum numbers) but does not produce scattering amplitudes, decay rates, or a Lagrangian.
- Spin remains an open problem, as does the derivation of weak mixing parameters and neutrino masses.

The aim of this paper is to show that the five quantum numbers, taken together, provide a highly constrained and internally consistent encoding of the observed spectrum—not to claim that the Standard Model has been derived from topology.

8.5 Connection to quantum group invariants

The confinement factor n_q^q may have a deeper origin in the Adams operation ψ_q of the Rosso–Jones formula [8]. Gorsky, Milekhin, and Sopenko [9] showed that HOMFLY torus knot invariants provide the entropic factor counting the degeneracy of instanton-W-boson web states, with the torus knot numbers (n, m) identified as instanton and electric charges. The positivity theorem of Etingof, Gorsky, and Losev [11] ensures that these invariants are genuine partition functions—they count states. Whether n_q^q is exactly the Adams operation evaluated at a specific root of unity remains an open question.

9 Conclusion

Five quantum numbers (p, q, m, n_q, f) provide a compact and highly constrained encoding of mass, charge, baryon number, and flavor across the particle spectrum considered here. The mass formula of [1] is consistent with established mathematics of torus knots [4, 5], vortex energetics [13], and quantum group invariants [8]. The new results of this paper—charge from framing and the genus-based structural criterion—extend the scheme beyond mass to encompass discrete quantum numbers, and provide additional structural support for the τ stealth baryon hypothesis of [1].

The separation of leptonic and hadronic states involves both topology and geometry: genus and constituent count jointly classify the knot, while the distinction between topological charge (framing $\rightarrow$ isospin) and geometric charge (circulation $\rightarrow$ lepton number) reflects the dual nature of the framework. Particle decays admit a natural interpretation as Dehn surgery operations, with neutrinos carrying the topology-change data—an interpretive correspondence whose dynamical content remains to be developed.

Whether these structural regularities reflect an underlying physical mechanism or an emergent mathematical organisation of the data is a question for further investigation. The experimental creation of torus knot wavefunctions in spinor Bose–Einstein condensates [16] may provide an independent avenue for testing aspects of this framework.

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Data and Code Availability

All analysis code is available at <https://github.com/JimGalasyn/null-worldtube>. The Jones polynomial analysis, charge verification, and surgery network calculations can be reproduced by running:

```
python3 analysis/nwt_charge_from_framing.py
python3 analysis/nwt_jones_polynomial.py
python3 analysis/nwt_surgery_network.py
```